

RESULTS

ICVGIP 2026 — Submission 485

Glacial Lake Segmentation using YOLOv11m-seg and SAM2 with Uncertainty-Guided Refinement

Annexure: Revised Tables and Figures

Table 1: Dataset Distribution

| Dataset    | Source                                                   | Images | Ground Truth |
|------------|----------------------------------------------------------|--------|--------------|
| Training   | Roboflow glacial-lake datasets (training split)          | 333    | 333          |
| Validation | Roboflow glacial-lake datasets (validation split)        | 464    | 464          |
| Test       | Kaggle Glacial Lake Dataset (Sentinel-2, 10 m, Himalaya) | 1640   | 1640         |
| Total      | All Dataset combined                                     | 2437   | 2437         |

Table 2: Performance of YOLOv11 variants and component-wise ablation

| S.No. | Model / Configuration                       | IoU    | Dice/F1 | Prec.  | Recall | Spec.  | Pix. Acc. | Bal. Acc. | Params |
|-------|---------------------------------------------|--------|---------|--------|--------|--------|-----------|-----------|--------|
| 1     | YOLOv11n-seg                                | 0.7511 | 0.8579  | 0.8950 | 0.8237 | 0.9877 | 0.9737    | 0.9067    | 2.9M   |
| 2     | YOLOv11s-seg                                | 0.7117 | 0.8315  | 0.8172 | 0.7605 | 0.8927 | 0.9703    | 0.8766    | 10.1M  |
| 3     | YOLOv11m-seg                                | 0.7021 | 0.8250  | 0.8368 | 0.7370 | 0.9947 | 0.9698    | 0.8659    | 22.4M  |
| 4     | YOLOv11l-seg                                | 0.7221 | 0.8386  | 0.8776 | 0.7794 | 0.9415 | 0.9711    | 0.8855    | 27.6M  |
| 5     | YOLOv11x-seg                                | 0.6978 | 0.8220  | 0.8627 | 0.7477 | 0.8424 | 0.9687    | 0.8700    | 62.1M  |
| 6     | YOLOv11m-seg + SAM2                         | 0.7600 | 0.8150  | 0.7520 | 0.8154 | 0.9868 | 0.9804    | 0.8991    | 103.2M |
| 7     | + Boundary-aware box                        | 0.6459 | 0.7849  | 0.7764 | 0.7936 | 0.9651 | 0.9423    | 0.8793    | 103.2M |
| 8     | + Boundary-aware box and points             | 0.6804 | 0.8098  | 0.8024 | 0.8174 | 0.9692 | 0.9491    | 0.8933    | 103.2M |
| 9     | + Random clicks (same number)               | 0.7092 | 0.8299  | 0.8499 | 0.8108 | 0.9781 | 0.9559    | 0.8944    | 103.2M |
| 10    | + Distance-based clicks (same number)       | 0.7049 | 0.8269  | 0.8436 | 0.8109 | 0.9770 | 0.9550    | 0.8940    | 103.2M |
| 11    | + Uncertainty-guided correction (proposed)  | 0.7876 | 0.8222  | 0.7636 | 0.8126 | 0.9867 | 0.9802    | 0.8997    | 103.2M |
| 12    | Full framework (box, points and correction) | 0.7876 | 0.8222  | 0.7636 | 0.8126 | 0.9867 | 0.9802    | 0.8997    | 103.2M |

Table 3: Comparison with other methods

| Dataset                                                                                                 | Method              | IoU    | Dice/F1 | Precision | Recall |
|---------------------------------------------------------------------------------------------------------|---------------------|--------|---------|-----------|--------|
| <i>(a) Trained on the same data and tested on the same test set</i>                                     |                     |        |         |           |        |
| Kaggle test set (Sentinel-2)                                                                            | U-Net               | 0.5145 | 0.6795  | 0.6058    | 0.7735 |
| Kaggle test set (Sentinel-2)                                                                            | DeepLabV3+          | 0.6403 | 0.7807  | 0.8706    | 0.7077 |
| Kaggle test set (Sentinel-2)                                                                            | YOLOv11m-seg + SAM2 | 0.7600 | 0.8150  | 0.7520    | 0.8154 |
| Kaggle test set (Sentinel-2)                                                                            | Proposed framework  | 0.7876 | 0.8222  | 0.7636    | 0.8126 |
| <i>(b) Reported in other papers on their own datasets (for reference only, not directly comparable)</i> |                     |        |         |           |        |

| Dataset                 | Method       | IoU    | Dice/F1 | Precision | Recall |
|-------------------------|--------------|--------|---------|-----------|--------|
| Sentinel-1 & Sentinel-2 | U-ViT        | 0.808  | 0.894   | 0.902     | 0.887  |
| Sentinel-1 & Sentinel-2 | U-Net        | 0.655  | 0.792   | 0.814     | 0.771  |
| Sentinel-1 & Sentinel-2 | DeepLabV3+   | 0.662  | 0.797   | 0.820     | 0.775  |
| Sentinel-2              | YOLOv5-seg   | 0.7160 | 0.7900  | 0.8245    | 0.7600 |
| Sentinel-2              | Adaptive SAM | 0.7600 | 0.8073  | 0.8200    | 0.8113 |

Table 4: Performance after each iteration

| Iteration    | IoU    | Dice/F1 | Precision | Recall |
|--------------|--------|---------|-----------|--------|
| Initial SAM2 | 0.7157 | 0.8343  | 0.8583    | 0.8116 |
| 1            | 0.7142 | 0.8333  | 0.8572    | 0.8107 |
| 2            | 0.7180 | 0.8359  | 0.8629    | 0.8104 |
| 3            | 0.7184 | 0.8361  | 0.8623    | 0.8115 |
| 4            | 0.7876 | 0.8222  | 0.7636    | 0.8126 |
| 5            | 0.7876 | 0.8222  | 0.7636    | 0.8126 |

Table 5: Validation of the uncertainty map against SAM2 error pixels

| Uncertainty map              | AUROC |
|------------------------------|-------|
| Attention entropy (proposed) | 0.728 |
| Distance to boundary         | 0.693 |
| Predictive entropy           | 0.559 |

Table 6: IoU and Dice under challenging conditions

| Condition                 | Images | IoU (YOLOv11m-seg + SAM2) | Dice (YOLOv11m-seg + SAM2) | IoU (Proposed Framework) | Dice (Proposed Framework) |
|---------------------------|--------|---------------------------|----------------------------|--------------------------|---------------------------|
| Small lakes               | 137    | 0.5774                    | 0.7321                     | 0.7024                   | 0.8283                    |
| Medium lakes              | 136    | 0.7263                    | 0.8415                     | 0.7618                   | 0.8686                    |
| Large lakes               | 137    | 0.7653                    | 0.8670                     | 0.8237                   | 0.9062                    |
| Regular boundary          | 137    | 0.8869                    | 0.9401                     | 0.9126                   | 0.9580                    |
| Irregular boundary        | 136    | 0.7396                    | 0.8503                     | 0.7864                   | 0.8838                    |
| Highly irregular boundary | 137    | 0.4837                    | 0.6520                     | 0.6218                   | 0.7696                    |
| Single lake               | 60     | 0.9559                    | 0.9775                     | 0.9452                   | 0.9753                    |

| Condition                       | Images | IoU (YOLOv11m-seg + SAM2) | Dice (YOLOv11m-seg + SAM2) | IoU (Proposed Framework) | Dice (Proposed Framework) |
|---------------------------------|--------|---------------------------|----------------------------|--------------------------|---------------------------|
| Multiple lakes                  | 350    | 0.6794                    | 0.8091                     | 0.7425                   | 0.8554                    |
| Snow, ice or cloud surroundings | 280    | 0.7674                    | 0.8684                     | 0.8014                   | 0.8928                    |
| Low-contrast boundary           | 137    | 0.6836                    | 0.8121                     | 0.6967                   | 0.8251                    |
| Clear water                     | 78     | 0.5859                    | 0.7389                     | 0.8691                   | 0.9334                    |

**Table 7: Computational cost on an NVIDIA T4 GPU**

| Component                           | Value                                                   |
|-------------------------------------|---------------------------------------------------------|
| YOLOv11m-seg                        | 59.3 ms (123.3 GFLOPs)                                  |
| SAM2 image encoder (once per image) | 309.6 ms (646.8 GFLOPs)                                 |
| SAM2 mask decoder (per iteration)   | 19.7 ms (3.66 GFLOPs)                                   |
| Average corrective clicks per image | 0.91                                                    |
| YOLOv11m-seg + SAM2 (total)         | 408.9 ms per image                                      |
| <b>Proposed framework (total)</b>   | <b>420.8 ms per image (2.38 images/s), 774.6 GFLOPs</b> |
| Peak GPU memory                     | 1.37 GB                                                 |
| Parameters                          | 103.2M (no additional parameters)                       |

**Table 8: Comparison with related prompt-refinement approaches** (Page No. 8, Section 4.5)

| Approach                                     | Needs training | Needs user clicks | Iterative correction | Uses model uncertainty |
|----------------------------------------------|----------------|-------------------|----------------------|------------------------|
| RITM, SimpleClick (interactive segmentation) | Yes            | Yes               | Yes                  | No                     |
| FocSAM                                       | Yes            | Yes               | Yes                  | No                     |
| VRP-SAM                                      | Yes            | No                | No                   | No                     |
| Adaptive SAM                                 | Yes            | No                | No                   | No                     |
| RS-SAM (remote-sensing SAM adaptation)       | Yes            | No                | No                   | No                     |
| Detector-to-SAM pipeline (YOLO + SAM2)       | No             | No                | No                   | No                     |
| <b>Proposed framework</b>                    | <b>No</b>      | <b>No</b>         | <b>Yes</b>           | <b>Yes</b>             |

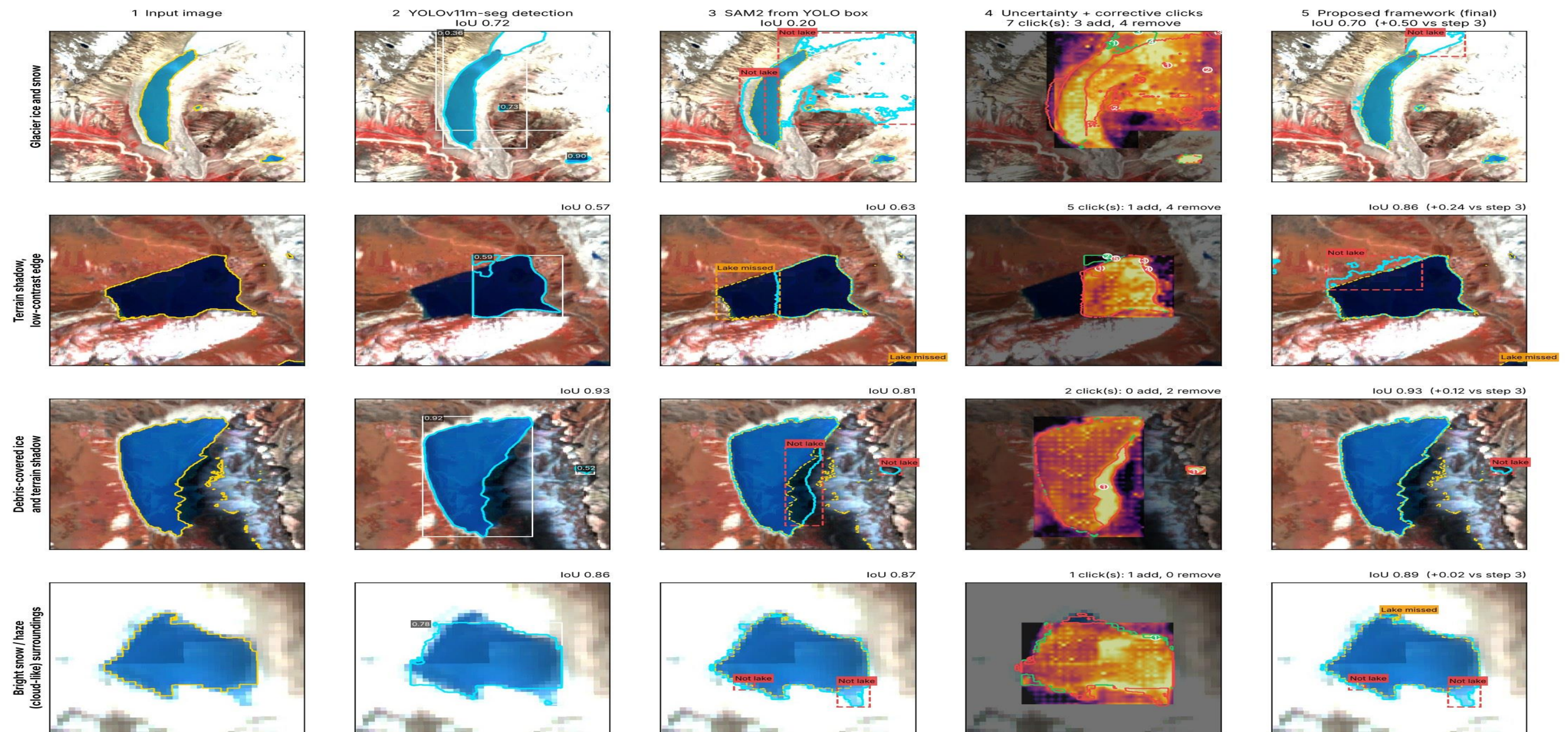
Table 9: Results for each Roboflow hard-case image

| Case                      | Split      | YOLOv11m-seg + SAM2 IoU | YOLOv11m-seg + SAM2 Dice | Proposed IoU | Proposed Dice | Proposed Precision | Proposed Recall | Clicks |
|---------------------------|------------|-------------------------|--------------------------|--------------|---------------|--------------------|-----------------|--------|
| Terrain shadow (Fig. 10a) | training   | 0.739                   | 0.850                    | 0.863        | 0.926         | 0.865              | 0.997           | 6      |
| Terrain shadow (Fig. 10b) | test       | 0.780                   | 0.876                    | 0.800        | 0.889         | 0.840              | 0.943           | 10     |
| Terrain shadow (Fig. 10c) | training   | 0.828                   | 0.906                    | 0.789        | 0.882         | 0.855              | 0.910           | 39     |
| Snow and ice (Fig. 11a)   | validation | 0.503                   | 0.669                    | 0.515        | 0.680         | 0.523              | 0.971           | 10     |
| Snow and ice (Fig. 11b)   | test       | 0.936                   | 0.967                    | 0.926        | 0.962         | 0.978              | 0.946           | 0      |
| Snow and ice (Fig. 11c)   | test       | 0.974                   | 0.987                    | 0.964        | 0.982         | 0.996              | 0.967           | 3      |
| Debris (Fig. 12a)         | validation | 0.716                   | 0.835                    | 0.718        | 0.836         | 0.895              | 0.784           | 14     |
| Debris (Fig. 12b)         | test       | 0.915                   | 0.956                    | 0.873        | 0.932         | 0.945              | 0.919           | 6      |
| Debris (Fig. 12c)         | validation | 0.686                   | 0.814                    | 0.730        | 0.844         | 0.818              | 0.872           | 14     |
| Mean (9 images)           |            | 0.786                   | 0.873                    | 0.797        | 0.881         | 0.857              | 0.923           | 11.3   |

Table 10: Comparison of primary and alternative data-split strategies and segmentation performance

| Data-Split Strategy           | Training    | Validation  | Test         | IoU    | Dice/F1 | Precision | Recall |
|-------------------------------|-------------|-------------|--------------|--------|---------|-----------|--------|
| Primary split (used in study) | 333 (13.7%) | 464 (19.0%) | 1640 (67.3%) | 0.7876 | 0.8222  | 0.7636    | 0.8126 |

| Data-Split Strategy          | Training     | Validation  | Test        | IoU    | Dice/F1 | Precision | Recall |
|------------------------------|--------------|-------------|-------------|--------|---------|-----------|--------|
| Alternative split (70:15:15) | 1706 (70.0%) | 366 (15.0%) | 365 (15.0%) | 0.7209 | 0.8378  | 0.8660    | 0.8114 |



Step 4 background: SAM2 attention entropy inside each YOLO box (contrast-stretched; dark = certain, bright = uncertain). Numbers in the clicks give the iteration.

— Ground-truth boundary     YOLO box (confidence)    — D-: SAM2 lake, YOLO background    ● Negative click (n = iteration)  
— Predicted boundary    — D+: YOLO lake, SAM2 background    ● Positive click (n = iteration)

Fig. 1: Step-by-step correction under challenging conditions

Glacier ice and snow

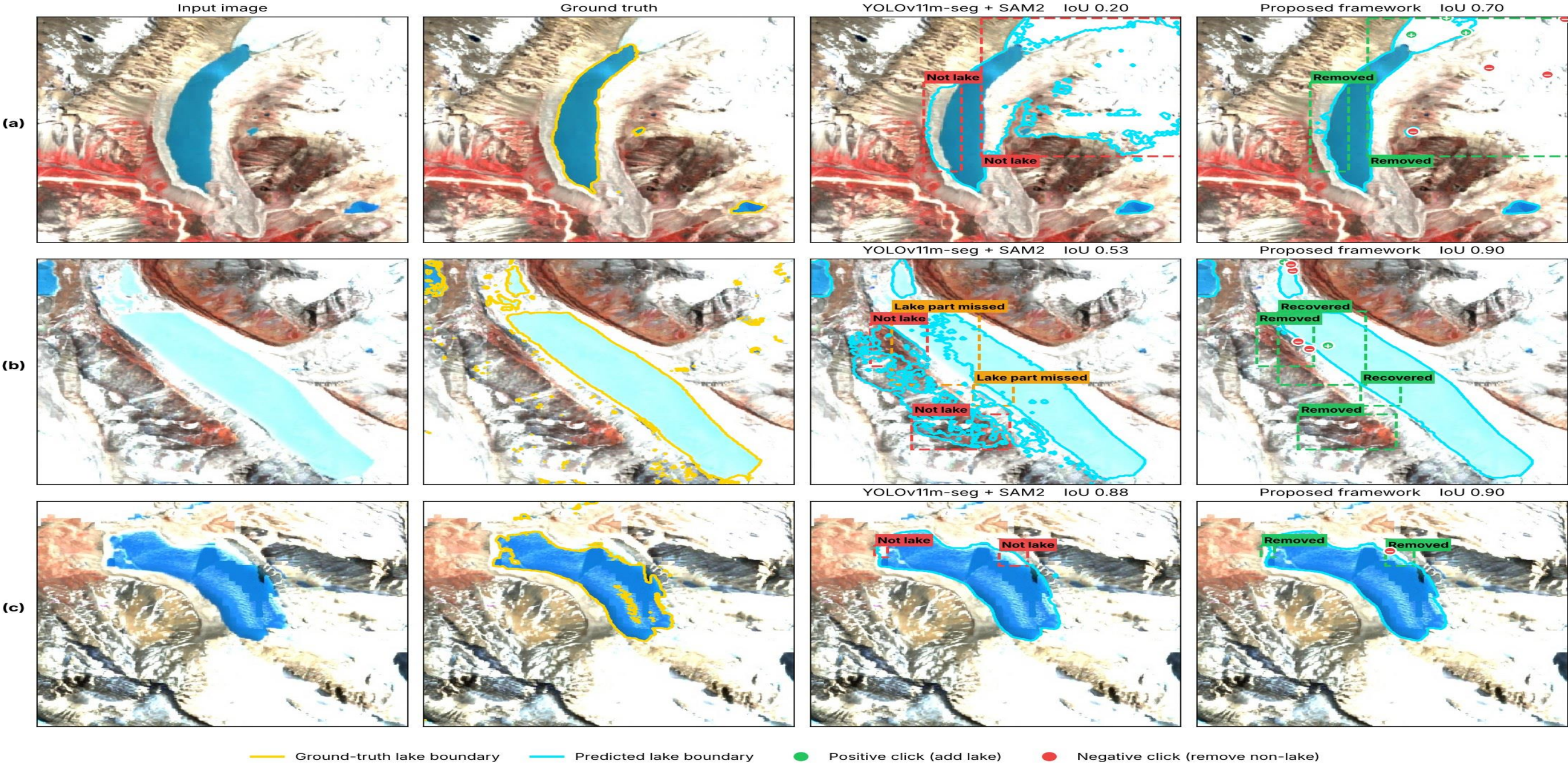


Fig. 2: Glacier ice and snow, Kaggle test images

### Bright cloud / haze-like surroundings

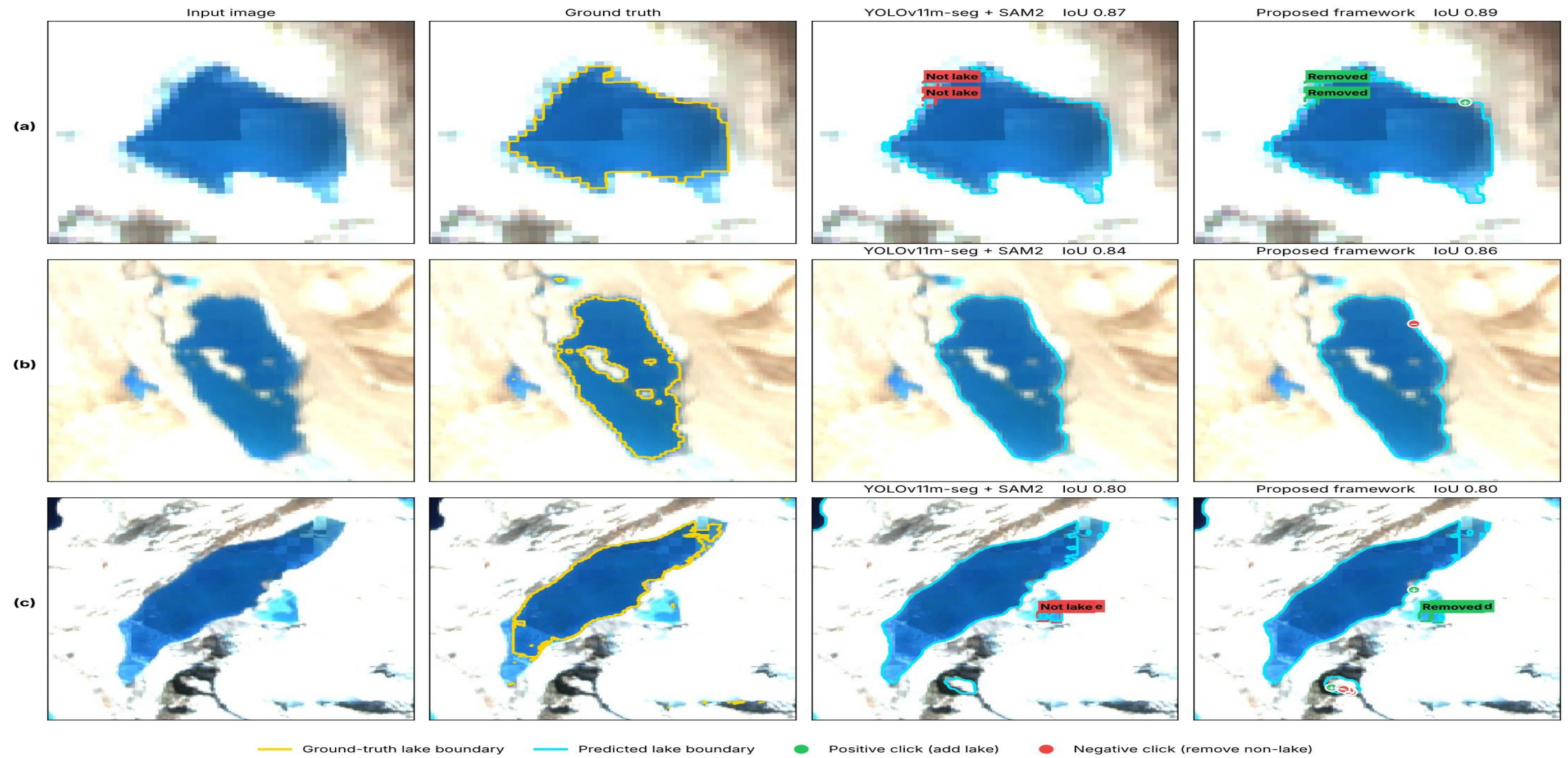


Fig. 3: Bright cloud- or haze-like surroundings, Kaggle test images

Low-contrast lake boundary and lake water missing from the ground truth

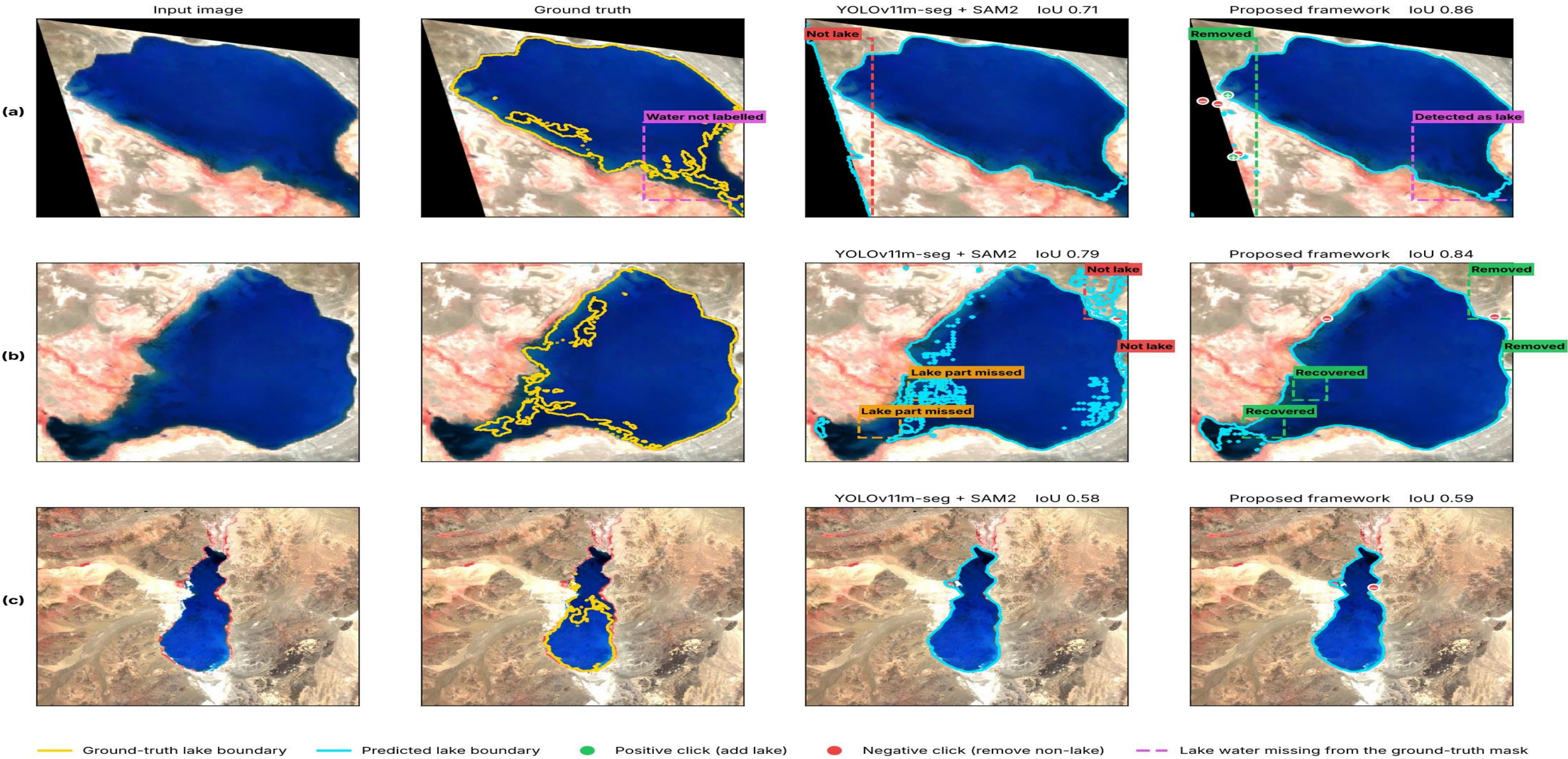


Fig. 4: Low-contrast lake edges and lake water missing from the ground truth, Kaggle test images

Hard case: terrain shadow (Roboflow dataset)

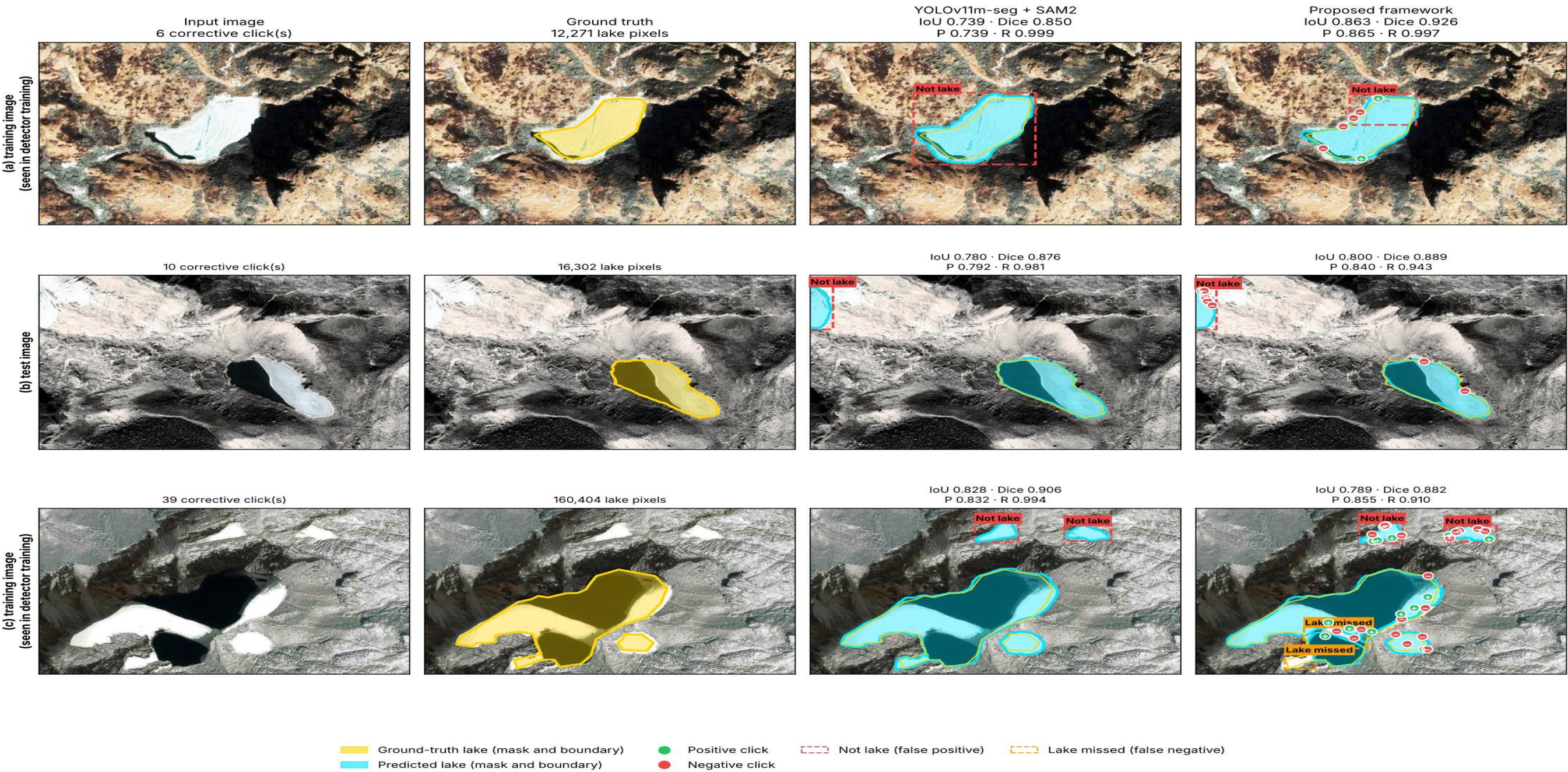


Fig. 5: Terrain shadow, Roboflow images

Hard case: snow and ice cover (Roboflow dataset)

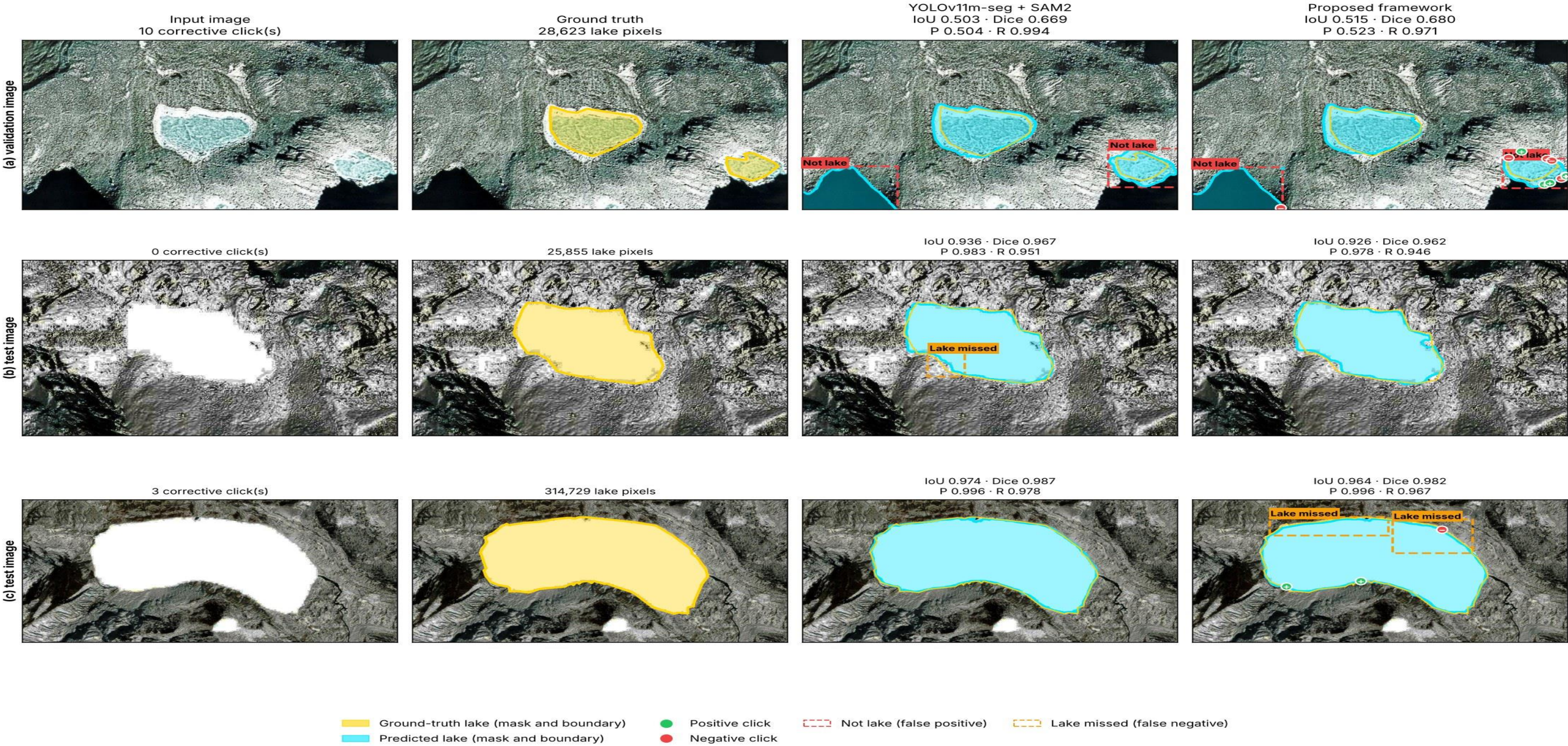


Fig. 6: Snow and ice cover, Roboflow images

Hard case: debris-covered surroundings (Roboflow dataset)

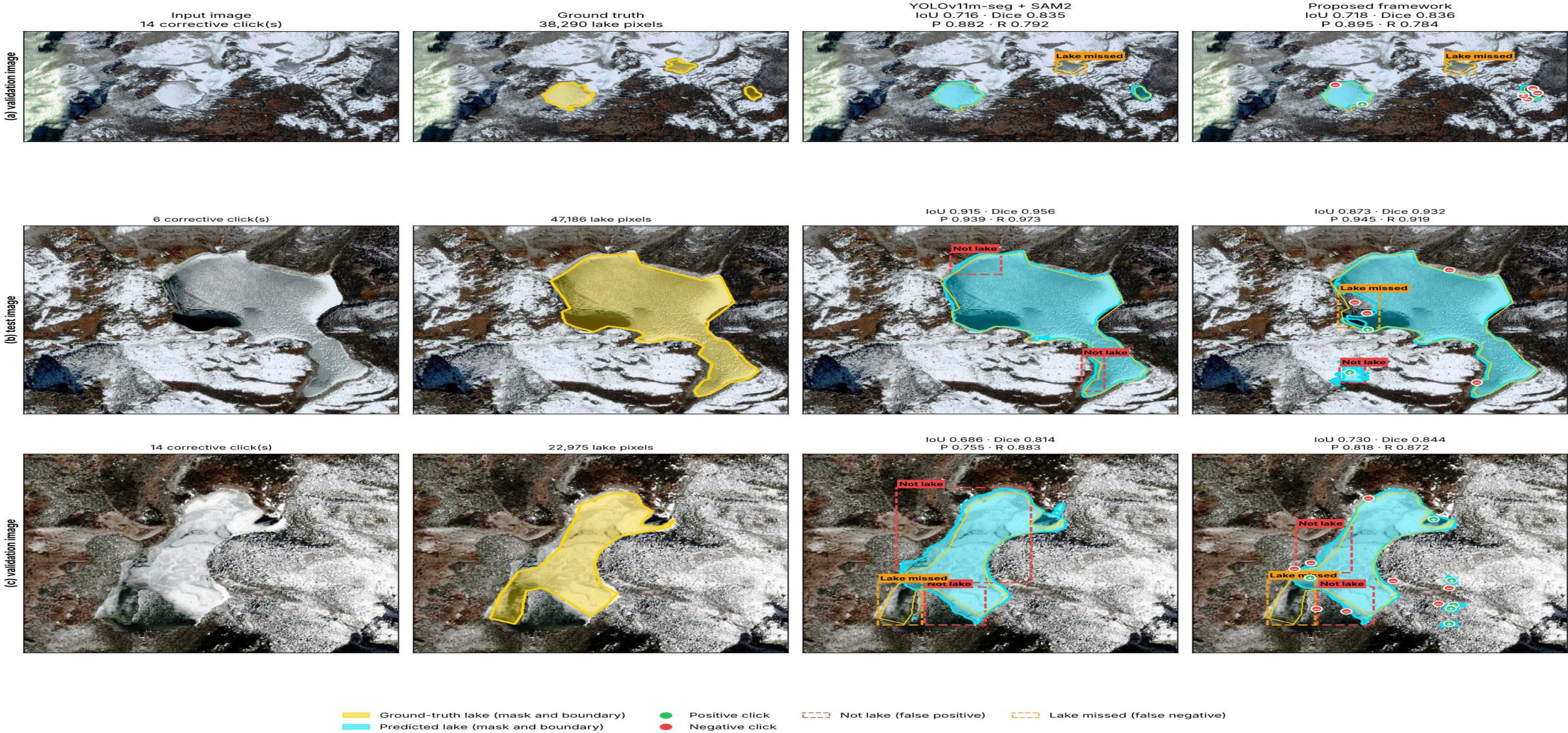


Fig. 7: Debris-covered surroundings, Roboflow images

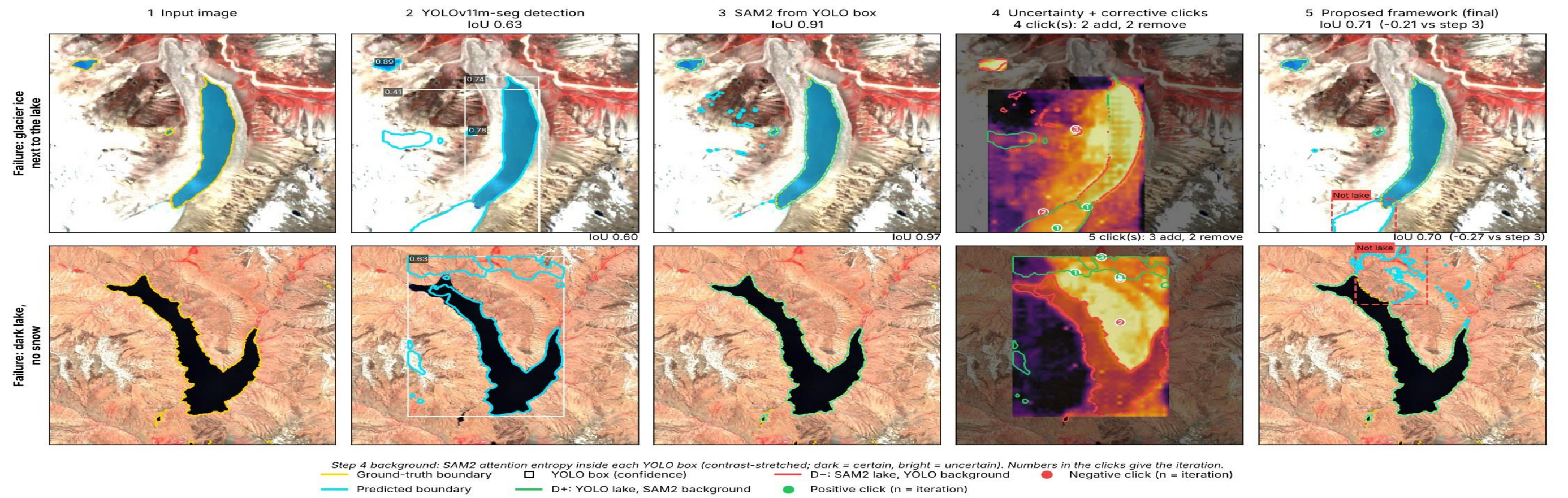


Fig. 8: Cases where the correction fails

Lake missed by YOLOv11m-seg but segmented by SAM2

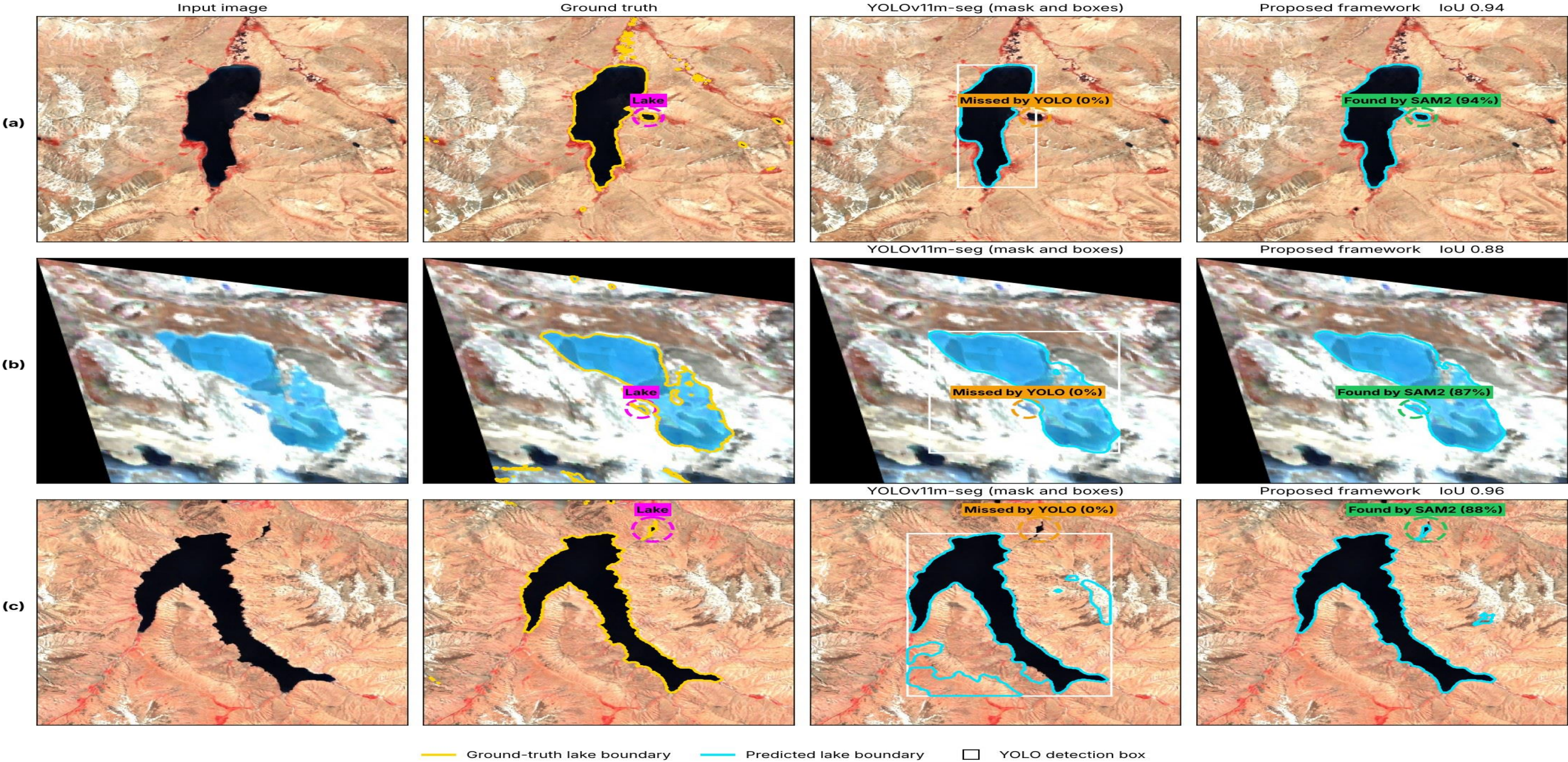


Fig. 9: Lakes missed by YOLOv11m-seg but found by SAM2